

DP Computer Science: A3.2.2 Entity Relationship Diagrams

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Understand the **significance of ERDs** in database design
- Identify and describe **relationships between data entities**
- Apply **cardinality** and **modality** to define relationships in ERDs
- Use **Chen's** and **Crow's Foot** notation to create ERDs

Key Vocabulary

Term	Definition
Entity	A real-world object or concept about which data is stored (e.g., Student, Course, Order)
Attribute	A property or characteristic of an entity (e.g., Student has Name, Age)
Relationship	An association between two or more entities (e.g., Student enrolls in Course)
Cardinality	The maximum number of occurrences in a relationship (1 or Many)
Modality	The minimum number of occurrences in a relationship (0 or 1)
ERD (Entity Relationship Diagram)	A visual diagram used to design databases, showing entities, attributes, and relationships
Chen notation	An ERD notation that uses diamonds for relationships and specific symbols for cardinality
Crow's Foot notation	An ERD notation that uses "crow's feet" symbols to represent cardinality
Primary key	A unique identifier for each record in a table
Foreign key	A field in one table that refers to the primary key of another table

IB ATL Skills

Skill	Description
Thinking	Critical thinking (analysing data relationships); creative thinking (designing ERDs)

Skill	Description
Communication	Visual communication (using ERD notation); explaining complex concepts
Self-Management	Organisation (structuring database designs)

2. Significance of ERDs in Database Design

What is an ERD?

An Entity Relationship Diagram (ERD) is a visual tool used to design databases. It illustrates the entities in a system and the relationships between them.

Why Use ERDs?

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WHY USE ERDS?

CLARITY

- Provides a clear visual representation of the data structure
- Helps stakeholders understand the design

EFFICIENCY

- Identifies potential problems early
- Reduces redundancy and improves data integrity

TAILORED DESIGN

- Can be adapted to any application
- Ensures the database meets specific requirements

Everyday Examples of Databases

Example	Entities	Relationships
Library	Books, Members, Borrowing	Member borrows Book
School	Students, Courses, Teachers	Student enrolls in Course, Teacher teaches Course
Online Store	Customers, Orders, Products	Customer places Order, Order contains Product

3. Entities and Attributes

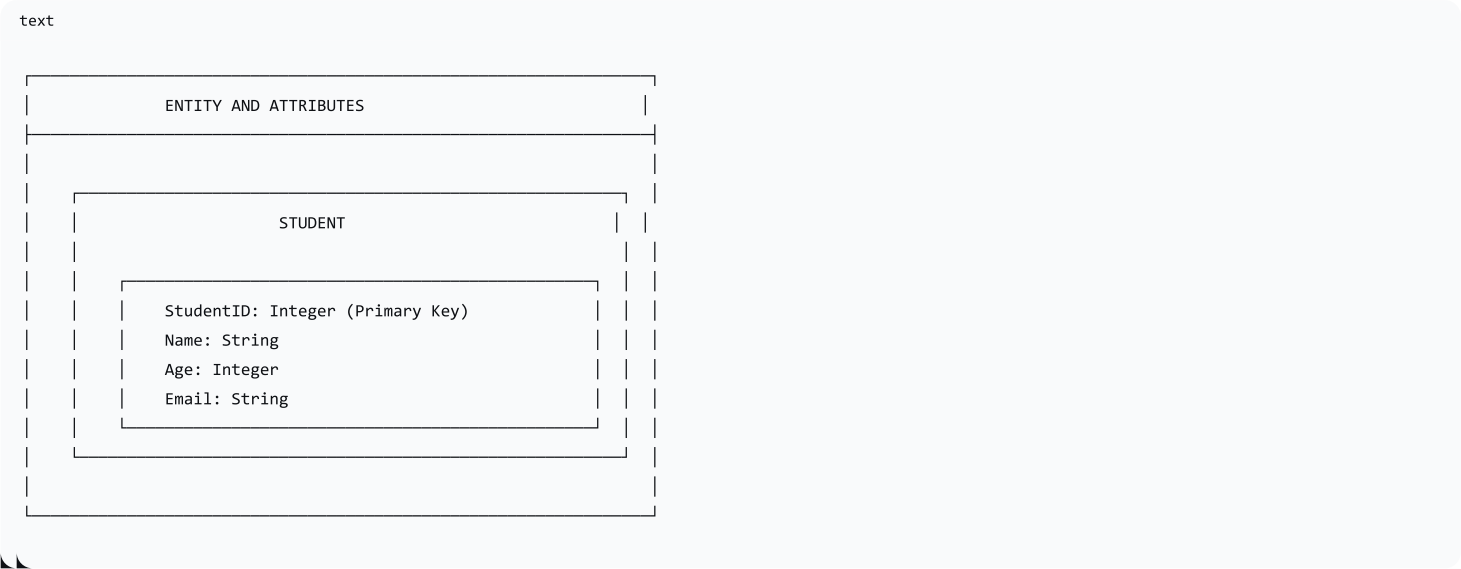
Entities

| An entity is a real-world object or concept about which data is stored.

Entity	Description	Example
Student	A person enrolled in a school	StudentID, Name, Age
Course	A class or subject	CourseID, Title, Credits
Order	A purchase transaction	OrderID, Date, Total

Attributes

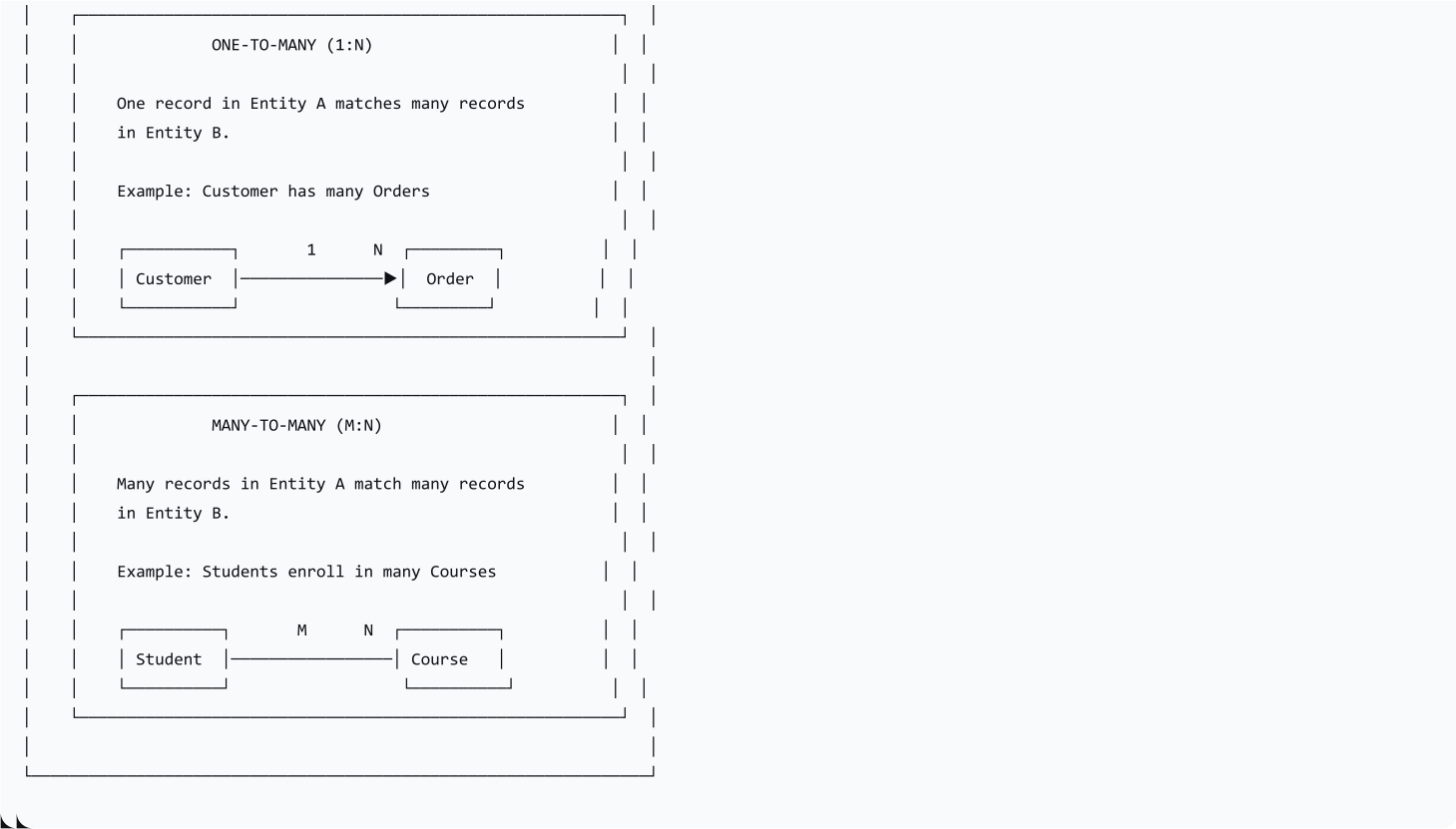
| An attribute is a property or characteristic of an entity.



4. Types of Relationships

The Three Relationship Types





5. Cardinality and Modality

Cardinality

Cardinality defines the maximum number of occurrences in a relationship.

Symbol	Meaning	Description
1	One	At most one occurrence
M	Many	Multiple occurrences possible

Modality

Modality defines the minimum number of occurrences in a relationship.

Symbol	Meaning	Description
0	Optional	The relationship is optional (minimum zero)
1	Mandatory	The relationship is required (minimum one)

Examples

Relationship	Cardinality	Modality	Meaning
Customer → Order	Many	0	A customer may have many orders, or none
Order → Customer	One	1	An order must belong to exactly one customer

Relationship	Cardinality	Modality	Meaning
Student → Course (Enrollment)	Many	0	A student may be enrolled in many courses, or none
Course → Student (Enrollment)	Many	0	A course may have many students, or none

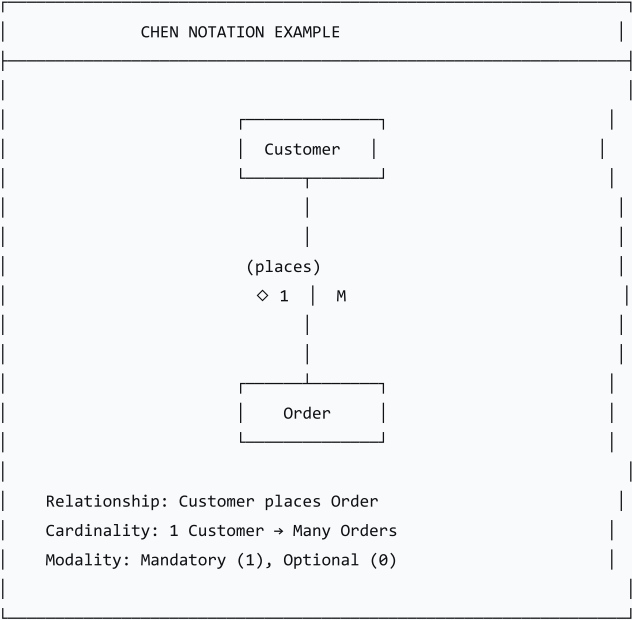
6. Chen Notation

Chen Notation Symbols

Element	Symbol	Description
Entity	Rectangle 	Represents an entity
Attribute	Oval 	Represents an attribute
Relationship	Diamond 	Represents a relationship
Cardinality		Maximum occurrences
Modality		Minimum occurrences

Chen Notation Example

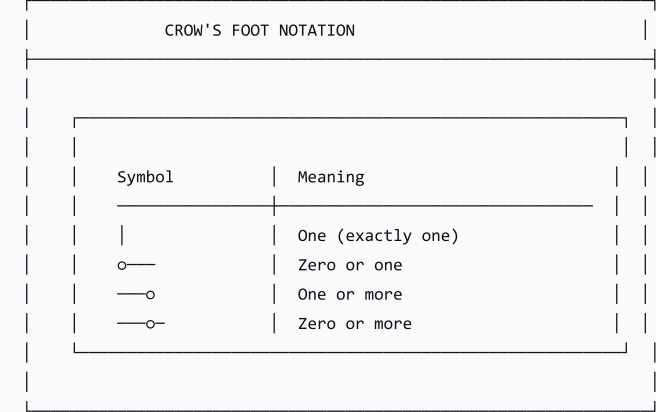
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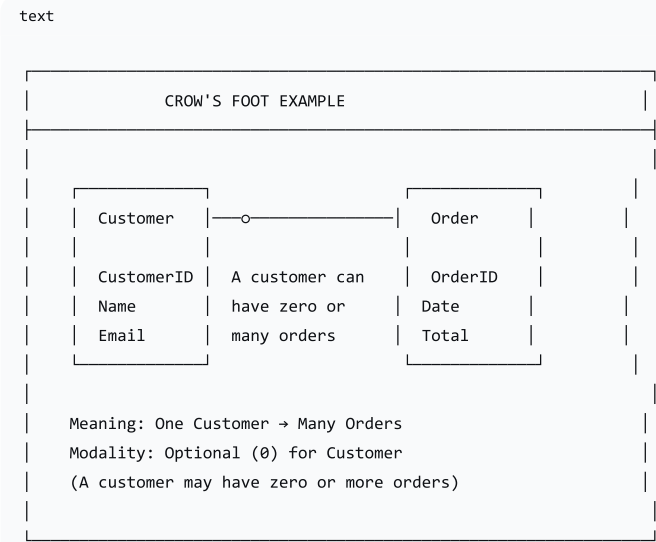
7. Crow's Foot Notation

Crow's Foot Symbols

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Crow's Foot Notation Example



8. Chen vs. Crow's Foot Notation Comparison

Feature	Chen Notation	Crow's Foot Notation
Entity	Rectangle	Rectangle
Attribute	Oval	None (usually omitted)
Relationship	Diamond	Line between entities
Cardinality	1, M next to diamond	Crow's foot symbols on line
Modality	0, 1 next to diamond	Symbols on line
Readability	Very clear for relationships	Very clear for cardinality
Complexity	Can become complex	Simpler and cleaner
Common Use	Academic/Conceptual	Industry/Implementation

9. Activity: ERD Design Practice

Scenario 1: School Database

Problem: A school needs a database to track students and the courses they enroll in. Each student can enroll in many courses, and each course can have many students.

Entity	Attributes
Student	StudentID (PK), Name, Age, Email
Course	CourseID (PK), Title, Credits

Relationship: Student enrolls in Course (Many-to-Many)

Solution:

- Create a junction table: **Enrollment**
- Enrollment: EnrollmentID (PK), StudentID (FK), CourseID (FK), Grade

Scenario 2: Library Database

Problem: A library needs a database to track books and the members who borrow them.

Entity	Attributes
Book	BookID (PK), Title, Author, ISBN
Member	MemberID (PK), Name, Address, Phone

Relationship: Member borrows Book (Many-to-Many)

Solution:

- Create a junction table: **Borrowing**
- Borrowing: BorrowID (PK), MemberID (FK), BookID (FK), DateBorrowed, DateReturned

Scenario 3: Online Store

Problem: An online store needs to track customers, orders, and products.

Entity	Attributes
Customer	CustomerID (PK), Name, Email, Address
Order	OrderID (PK), Date, Total, CustomerID (FK)
Product	ProductID (PK), Name, Price, Stock

Relationships:

- Customer places Order (One-to-Many)
- Order contains Product (Many-to-Many)

Solution:

- Customer → Order: One-to-Many
- Order → Product: Many-to-Many (use OrderDetail junction table)

10. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	10 min	Class discussion on everyday databases; significance of ERDs
Entities & Relationships	15 min	Identifying entities, attributes, and relationship types
Cardinality & Modality	15 min	Applying cardinality and modality to relationships
ERD Notation	10 min	Chen and Crow's Foot notation; teacher-led examples
Practice	10 min	Students design ERDs for given scenarios

11. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Worksheet	Evaluate ability to create and interpret ERDs
Exit Ticket	Gauge understanding and identify areas needing clarification

12. Differentiation

Level	Strategy
Support	Provide visual aids and step-by-step examples
Challenge	Design ERDs for complex multi-entity systems

13. Resources

- Presentation (Slide Deck)
- eBook (A3.2.2 with examples)
- Worksheet: ERD design and interpretation
- Quizlet (A3.2.2 vocabulary flashcards)

14. Summary: Key Takeaways

Concept	Key Point
Entity	A real-world object (e.g., Student, Order)
Attribute	A property of an entity (e.g., Name, Date)
Relationship	Association between entities (e.g., enrolls, places)

Concept	Key Point
Cardinality	Maximum occurrences (1 or Many)
Modality	Minimum occurrences (0 or 1)
Chen Notation	Uses diamonds for relationships
Crow's Foot	Uses symbols for cardinality (industry standard)

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KEY REMINDERS	
✓	ERDs are visual tools for designing databases
✓	Entities are real-world objects
✓	Attributes are properties of entities
✓	Relationships connect entities
✓	Cardinality = maximum occurrences
✓	Modality = minimum occurrences
✓	Chen = diamonds for relationships
✓	Crow's Foot = industry standard

“ERDs are the blueprint of a database—they define entities, their attributes, and how they are related before any tables are created.”